Project Report

E-Commerce Management System

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Course: B.Tech (CSE)

Technology Stack: Java, JDBC, Maven, MySQL, Eclipse

1. INTRODUCTION

E-commerce refers to the buying and selling of goods and services through electronic means, mainly the internet. In today's digital era, ecommerce has become an essential part of modern business. It provides customers with the convenience of shopping from anywhere at any time. Online platforms such as Amazon, Flipkart, and Meesho are popular examples of ecommerce systems.

The Ecommerce System project is designed to simulate the core operations of an online shopping platform. This project allows customers to register, browse products, add products to cart, place orders, and make payments. It also allows administrators to manage products, categories, and users.

The system is developed as a command-line application using Java. MySQL is used as the backend database, and JDBC is used to establish connectivity between Java and the database. Maven is used for dependency management and project build automation. The aim of this project is to understand software development life cycle, database design, and Java database connectivity concepts.

This project provides hands-on experience in creating a real-life application that interacts with a database, processes user requests, and manages data efficiently. The Ecommerce System demonstrates how data is stored, retrieved, and updated dynamically in an online shopping environment.

2. OBJECTIVES

The main objective of the Ecommerce System project is to develop a functional online shopping platform using Java and MySQL. The specific objectives of this project are as follows:

1. To design a database that stores customer, product, category, order, payment, and review information.

2. To implement customer registration and login functionality.

3. To allow customers to browse products and view product details.

4. To provide cart management features such as adding, updating, and removing products.

5. To enable customers to place orders and make payments.

6. To allow administrators to manage products, categories, and customers.

7. To ensure secure storage and retrieval of data using MySQL.

8. To understand and implement JDBC for database connectivity.

9. To follow modular programming and software engineering principles.

By achieving these objectives, the project successfully simulates a real ecommerce environment and strengthens understanding of Java programming and database integration.

3. FEASIBILITY STUDY

Feasibility study is an important step in software development that determines whether a project is practical and achievable. The Ecommerce System project has been analyzed under different feasibility aspects.

Technical Feasibility:

The project uses Java, MySQL, JDBC, Maven, and Eclipse, which are widely available and supported technologies. The required hardware and software resources are easily accessible. Therefore, the project is technically feasible.

Operational Feasibility:

The system is designed to be user-friendly even though it is command-line based. Users can easily perform operations by following simple menu-driven instructions. Admin operations are also simplified. Hence, the system is operationally feasible.

Economic Feasibility:

All tools used in this project are open-source and free of cost. There is no requirement for paid software or expensive hardware. Thus, the project is economically feasible.

Schedule Feasibility:

The project can be completed within the academic time frame as it uses standard development tools and follows a structured design approach.

Hence, the Ecommerce System project is feasible in all aspects.

4. SYSTEM REQUIREMENT ANALYSIS

System requirement analysis defines the hardware and software requirements needed to run the project successfully.

Hardware Requirements:

The system requires a computer with an Intel i3 processor or above for smooth performance. A minimum of 4GB RAM is required to run Eclipse, MySQL, and Java applications simultaneously. At least 500GB of hard disk space is recommended to store software tools and database files. Standard input/output devices like keyboard and mouse are also required.

Software Requirements:

The project requires Windows 10 or 11 operating system. Java Development Kit (JDK 8 or above) is needed for developing and running Java programs. Eclipse IDE is used for writing and managing the source code. MySQL database server is used to store all project data. JDBC is used to connect Java with MySQL. Maven is used for managing project dependencies and build automation.

These requirements ensure that the project runs smoothly and efficiently without performance issues.

5. SYSTEM DESIGN

System design defines how the system works internally. The Ecommerce System follows a modular architecture where each module handles a specific task.

The Customer Module handles customer registration, login, and profile management. The Product Module manages product details such as name, description, price, and stock. The Category Module classifies products into different groups. The Cart Module allows customers to add and manage products before placing orders. The Order Module processes customer orders and maintains order status. The Payment Module handles payment details and transaction records. The Review Module allows customers to provide feedback on products. The Admin Module manages overall system operations.

The system uses a MySQL database to store all records. Java classes interact with database tables through JDBC. This modular approach makes the system easy to understand, maintain, and upgrade in the future.

6. DATABASE DESIGN

The database is the backbone of the Ecommerce System. It is designed using the relational model and consists of multiple interrelated tables.

The database name is ecommerce_db. The customers table stores customer details such as name, email, password, phone, and address. The categories table stores different product categories. The products table stores product information and is linked to categories through foreign keys. The orders table stores customer orders. The order_items table stores details of products in each order. The cart_items table stores products added to customer carts. The payments table stores transaction details. The reviews table stores customer feedback on products. The admin_users table stores administrator login information.

Primary keys uniquely identify records, and foreign keys maintain relationships between tables. This design ensures data integrity and efficient data retrieval.

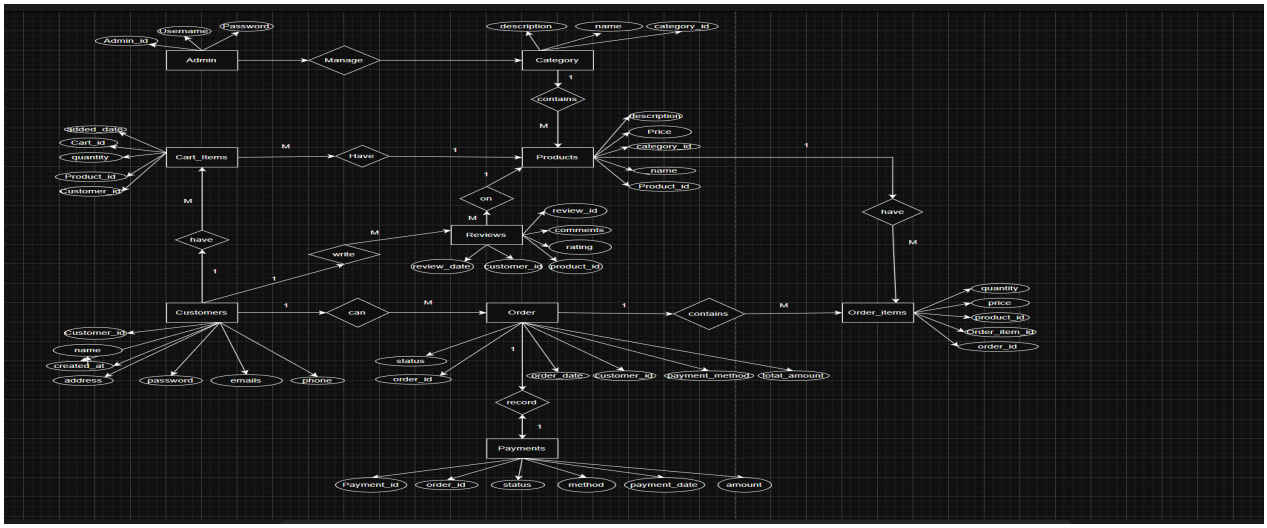
7. ENTITY RELATIONSHIP DIAGRAM

The Entity Relationship Diagram (ERD) represents the logical structure of the database.

Entities in the system include Customers, Categories, Products, Orders, Order_Items, Cart_Items, Payments, Reviews, and Admin_Users. Each entity has attributes such as customer_id, product_id, order_id, etc.

Relationships:
A customer places many orders. Each order contains multiple order items. Each order item refers to a single product. A product belongs to a category. A customer can add many products to the cart. Each order has one payment record. A customer can give multiple reviews on different products.

The ER diagram visually represents these entities and relationships, ensuring clear understanding of data flow and database connectivity.

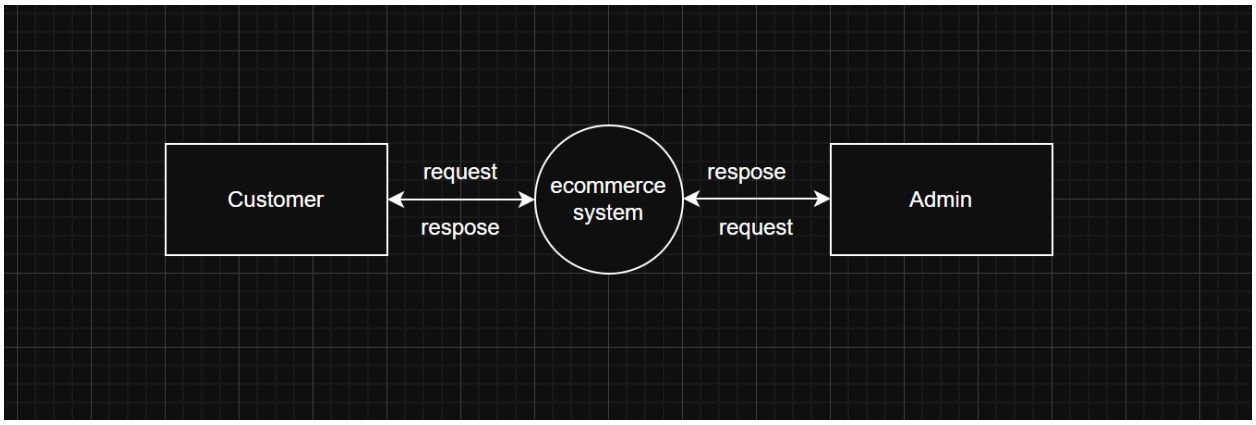


8. DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) shows how data moves through the system.

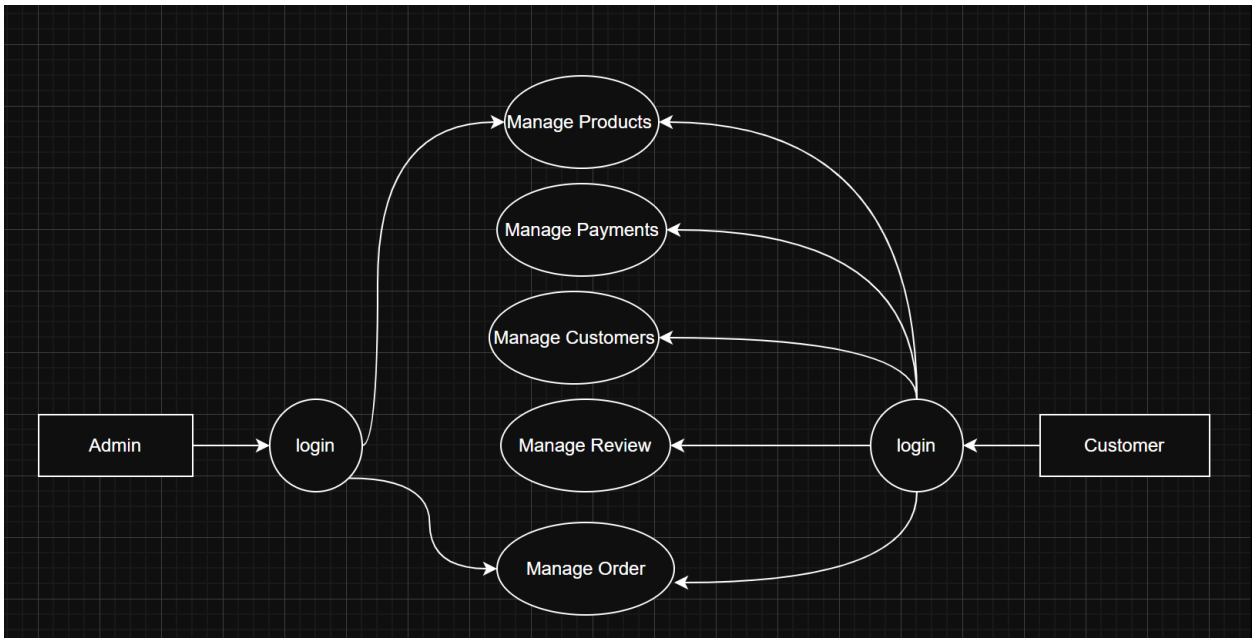
DFD Level 0 (Context Diagram):

At Level 0, the Ecommerce System is represented as a single process interacting with external entities such as Customer and Admin. Customers provide input such as login details, product selection, and payment information. Admin provides input to manage products and users. The system stores and retrieves data from the database.



DFD Level 1:

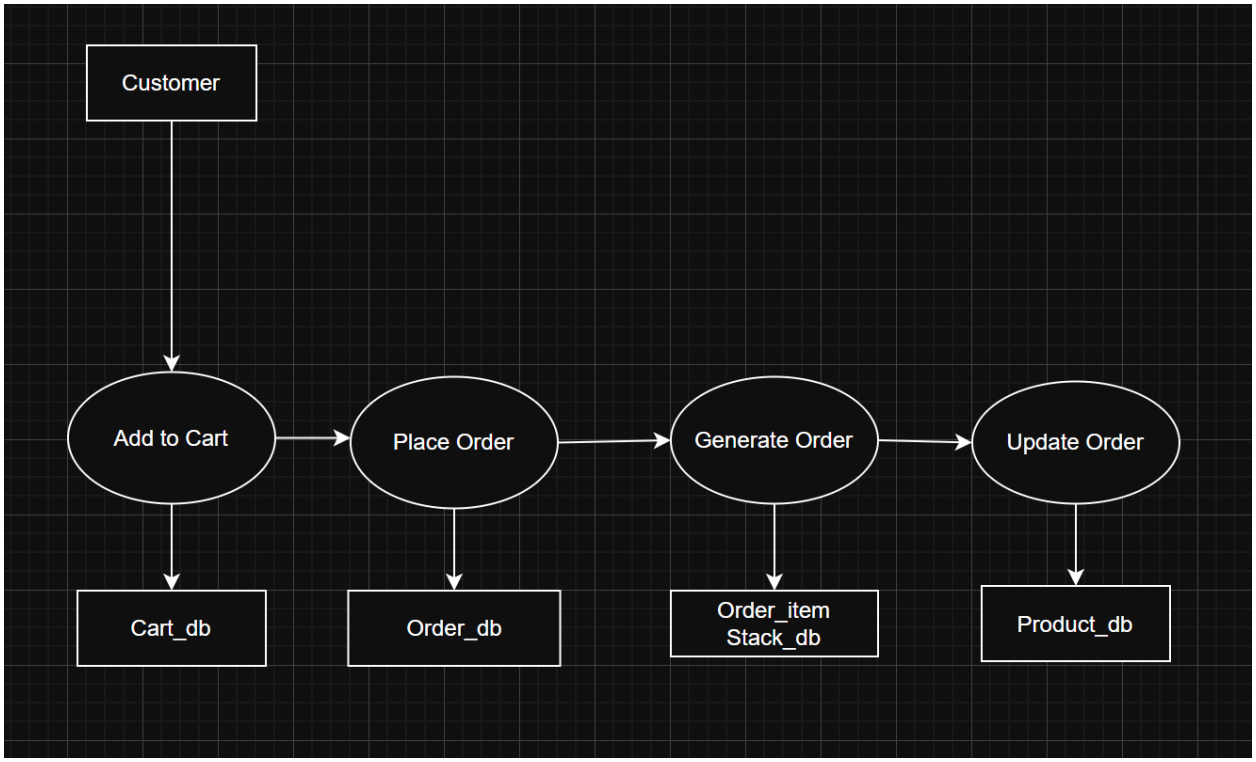
At Level 1, the system is divided into multiple processes such as User Management, Product Browsing, Cart Management, Order Processing, and Payment Processing. Each process interacts with specific database tables.



DFD Level 2:

At Level 2, the Order Processing module is broken down further into selecting products, adding to cart, placing order, generating order items, and recording payment.

The DFD helps in understanding how information flows within the system.



9. MODULE DESCRIPTION

Customer Module:
This module handles customer registration, login, and profile management.

Category Module:
This module manages different product categories.

Product Module:
This module handles product details such as name, description, price, and stock.

Cart Module:
This module allows customers to add, update, and remove products from the cart.

Order Module:
This module processes customer orders and stores order details.

Payment Module:
This module records payment information and transaction status.

Review Module:
This module allows customers to give feedback on products.

Admin Module:
This module manages products, categories, customers, and system data.

Each module is implemented using separate Java classes and interacts with the database through JDBC.

10. IMPLEMENTATION

The Ecommerce System is implemented using Java programming language. JDBC is used to establish a connection between Java and MySQL database. SQL queries are executed through Java to perform insert, update, delete, and retrieve operations.

The project follows a menu-driven command-line interface. When the program starts, the user is prompted to choose between customer and admin login. Based on the selection, appropriate options are displayed. Each functionality such as adding products, viewing cart, placing orders, and making payments is implemented in separate Java methods.

Maven is used to manage external libraries and simplify project build process. This implementation approach ensures clean, organized, and maintainable code.

11. TESTING

Testing is an important phase to verify that the system works correctly. Each module is tested separately to ensure correct functionality.

Customer registration and login testing ensures that users can create accounts and log in successfully. Product management testing ensures that products can be added, updated, and displayed correctly. Cart testing verifies that products can be added and removed from the cart. Order testing checks whether orders are created and stored properly. Payment testing verifies that payment records are saved in the database.

Database testing ensures data consistency and integrity. All test cases are executed successfully, confirming that the system performs as expected.

12. RESULT AND OUTPUT

The Ecommerce System successfully performs all required operations. Customers can register, browse products, manage cart, place orders, and make payments. Admin can manage products, categories, and users. All data is stored securely in MySQL database and retrieved accurately using JDBC.

The project output is displayed through command-line interface where users interact with menu options. Each operation provides confirmation messages indicating successful completion. The database reflects real-time updates after every operation.

The results confirm that the project meets all functional requirements.

13. CONCLUSION

The Ecommerce System project has been successfully developed using Java, JDBC, and MySQL. It fulfills the objectives of creating a basic online shopping simulation. The project demonstrates important concepts such as database design, SQL queries, Java programming, and JDBC connectivity.

This project provides practical experience in developing a real-world application. It also improves understanding of software development life cycle and modular programming approach. The system is reliable, efficient, and easy to use.

Thus, the Ecommerce System project is successfully completed and meets academic and technical requirements.

14. FUTURE SCOPE

The Ecommerce System can be further enhanced with additional features. It can be converted into a web-based application using technologies such as HTML, CSS, JavaScript, and JSP/Servlets. A graphical user interface can be added for better user experience. Online payment gateway integration can be implemented for real transactions. Advanced search and filtering options can be included. Email and OTP verification can improve security.

In the future, mobile application development and cloud database integration can also be implemented to make the system more scalable and modern.

15. BIBLIOGRAPHY

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